SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2013

BIOLOGY
Higher 1

Paper 1

Wednesday
28 August 2013

1 hour

Additional materials:
Optical Mark Sheet

INSTRUCTIONS TO CANDIDATES

Write you name, CG and index number in the spaces at the top of this page.

On the Optimal Mark Sheet, enter your name, subject title, test name, class. For your index number, enter your full NRIC number. Shade the corresponding lozenges on the OMS according to the instructions given by the invigilators.

There are forty questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A**, **B**, **C**, **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the OMS.

AT THE END OF THE EXAMINATION, HAND IN BOTH THE OMS AND QUESTION PAPER.

INFORMATION FOR CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done on the question paper.

This question paper consists of 19 printed pages.

1. A population of cells that produces secretory proteins is transferred into a medium containing only radioactively-labelled amino acids. In which of the following cell structures will radioactivity first be significantly detected?

A endoplasmic reticulum

B nucleus

C phagocytic vesicle

D secretory vesicle

2. **Fig. 2** below shows two lipid molecules, Lipid A and Lipid B.

Lipid A

Lipid B

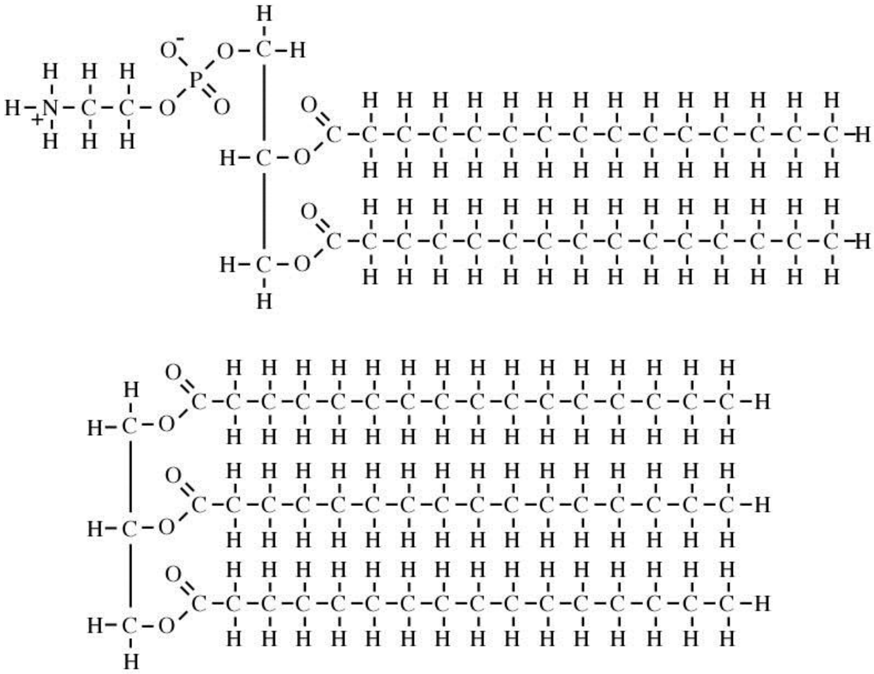


Fig. 2

Which of the following correctly indicates a similarity and difference between Lipids A and B?

	Similarity	Difference	
		Lipid A	**Lipid B**
A	Presence of strong ester bonds.	Presence of phosphodiester bond	Absence of phosphodiester bond
B	Hydrogen bonding between hydrocarbon chains	Presence of two ester bonds	Presence of three ester bonds
C	Presence of straight hydrocarbon chains	Amphipathic	Hydrophobic
D	Complete hydrolysis releases three water molecules	Contains phosphorus atom	Does not contain phosphorus atom

3. Which of the following correctly indicates a difference between collagen and haemoglobin?

	Collagen	**Haemoglobin**
A	Contains mostly hydrophobic amino acids	Contains mostly hydrophilic amino acids
B	Stabilised largely by numerous hydrogen bonds	Stabilised largely by numerous ionic and hydrogen bonds, and hydrophobic interactions
C	Has primary and secondary structures only	Has primary, secondary, tertiary and quaternary structures
D	Three subunits wind together to form triple helix	Each subunit has an active site which binds haem

4. Which of the following accounts for why starch and glycogen are good energy stores in living organisms?

A A large amount of energy is released when α(1→4) glycosidic bonds in starch and glycogen are hydrolysed.

B The occurrence of branch points every 26 to 30 residues in starch and glycogen provides many branch ends from which digestion can begin.

C The projection of hydroxyl groups into the helical structures of starch and glycogen causes both to be insoluble in water.

D Starch and glycogen are highly reactive macromolecules that can be rapidly broken down when required.

5. Which of the following does **not** describe the active site of an enzyme?

A The active site is usually a pocket comprising several amino acids.

B The active site binds with an inhibitor which has a complementary shape.

C In anabolic reactions, the active site brings substrates close together and catalyses bond formation.

D Beyond the optimum temperature, the active site is first denatured before the rest of the enzyme structure is denatured.

6. Which of the following regarding homologous chromosomes is correct?

A The X and Y chromosomes in human males are not homologous chromosomes.

B Homologous chromosomes are not present in a cell undergoing mitosis.

C Centromeres of homologous chromosomes occupy the same position that is approximately equidistant from both ends of the linear chromosomes.

D Homologous chromosomes usually carry the same alleles at corresponding gene loci.

7. **Fig. 7** below shows measurements during one mitotic cell cycle.

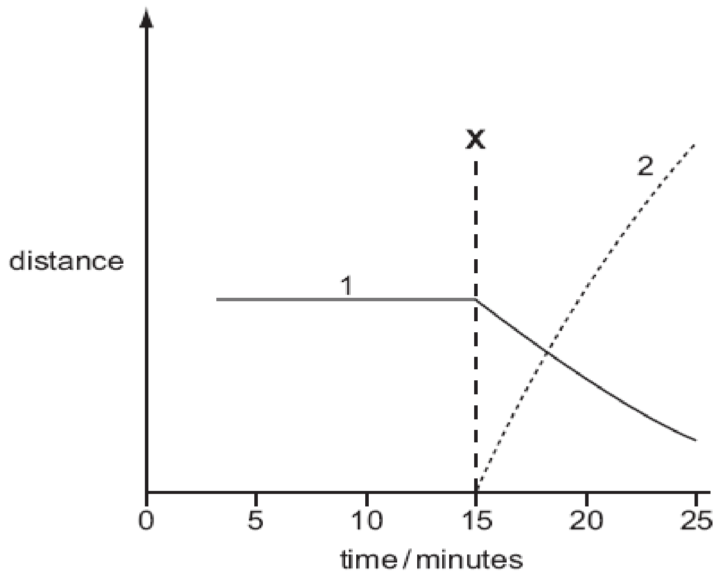


Fig. 7

Which stage of mitosis begins at **X** and which measurements are illustrated by curves 1 and 2?

	Stage beginning at X	**Distance between centromeres of chromosomes and poles of spindle**	**Distance between the centromeres of sister chromatids**
A	Metaphase	1	2
B	Metaphase	2	1
C	Anaphase	1	2
D	Anaphase	2	1

8. Which of the following does **not** explain how meiosis leads to genetic variation?

A At metaphase II, the orientation of a particular chromosome to the poles is not affected by the orientation of the other chromosomes along the equator.

B By the end of meiosis I, each daughter nucleus contains either the paternal or maternal copy of a chromosome.

C Homologous regions of DNA are exchanged within a bivalent after synapsis has occurred.

D The chromatids that are separating during anaphase II may be genetically different.

9. Several aspects of DNA replication are listed below:

1. Synthesis of complementary RNA primer
2. Base-pairing between DNA nucleotides
3. Removal of mismatched DNA nucleotides
4. Initiation only at origin of replication

Which of the above contributes directly to the accuracy of DNA replication?

A 1 and 3 only.

B 2 and 3 only.

C 3 and 4 only.

D 2, 3 and 4 only.

10. A simplified representation of a replication bubble is shown in **Fig. 10** below. Parental strands 1 and 2 and the growing daughter strands X and Y are indicated.

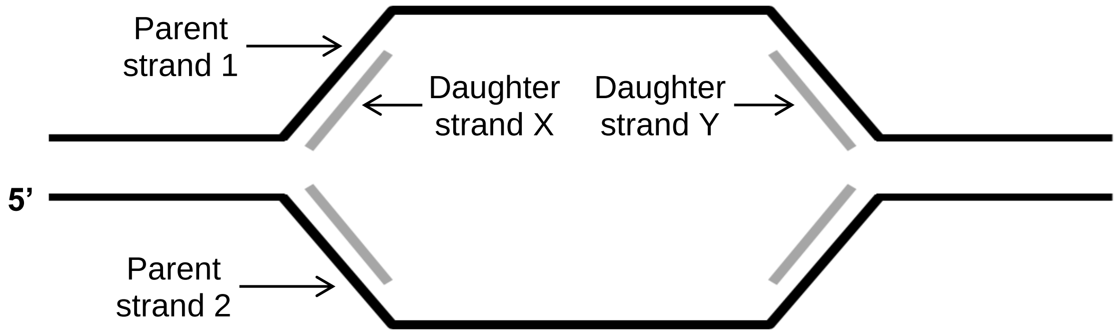


Fig. 10

Which of the following statements about the syntheses of Daughter strands X and Y is correct?

A Strands X and Y are synthesised away from their respective replication forks.

B Strand X is synthesised continuously while Strand X is synthesised in the form of Okazaki fragments.

C Strand X is synthesised in the 5' → 3' direction while Strand Y is synthesised in the 3' → 5' direction.

D DNA ligase will eventually catalyse the fusion of Strand X with Strand Y.

11. Translation of the information on an mRNA into the amino acid sequence of a polypeptide chain is an energy-requiring process, involving further processes such as:

1. Amino acid activation
2. Assembly of translation initiation complex
3. Attachment of amino acid to growing polypeptide chain
4. Translocation of ribosome along mRNA

In which of the above processes is **energy in the form of ATP** required?

A 1 only.

B 1 and 3 only.

C 1, 3 and 4 only.

D All of the above.

12. The first five DNA triplets that code for a particular protein is shown below:

3’ CAC GGA AGC CCA GAA 5’

The genetic information in the sequence above is eventually converted into a specific amino acid sequence according to **Fig. 12** below.

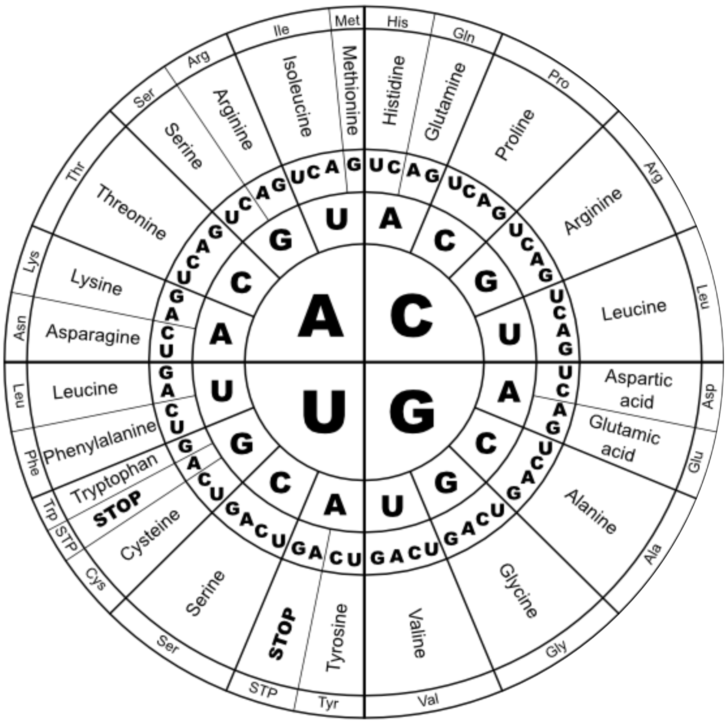


Fig. 12

With the aid of **Fig. 12**, what is the sequence of the protein encoded by the DNA sequence above?

A His – Gly – Ser – Pro – Glu

B Val – Pro – Ser – Gly – Leu

C Lys – Thr – Arg – Arg – His

D Phe – Try – Ala – Ser – Val

13. Which of the following is true of cancers?

A A cancer cell will always eventually develop into a tumour.

B Cancer cells are able to divide uncontrollably despite normal regulation of cell cycle checkpoints.

C A cell that has a copy of the *p53* tumour suppressor gene inactivated can be considered to be cancerous.

D When a copy of the *ras* proto-oncogene is activated into an oncogene in a normal cell, cancer immediately develops.

14. Two types of plumage (feathers) exist in chickens: hen-feathering and cock-feathering. The type of plumage results from the expression of an autosomal gene locus that carries two alleles:

- Allele **H** (hen-feathering) is dominant in both female and male chickens.
- Allele **h** (cock-feathering) is recessive, and the phenotype is expressed only in male chickens.

Which of the following would be the result of a mating between a cock-feathering male chicken, and a hen-feathering female chicken which is heterozygous at the gene locus?

A The probability of a male offspring being cock-feathering is 50% and that of a female offspring being hen-feathering is 50%.

B The probability of a male offspring being cock-feathering is 50% and that of a female offspring being hen-feathering is 100%.

C The probability of a male offspring being cock-feathering is 100% and that of a female offspring being hen-feathering is 50%.

D The probability of a male offspring being cock-feathering is 100% and that of a female offspring being hen-feathering is 100%.

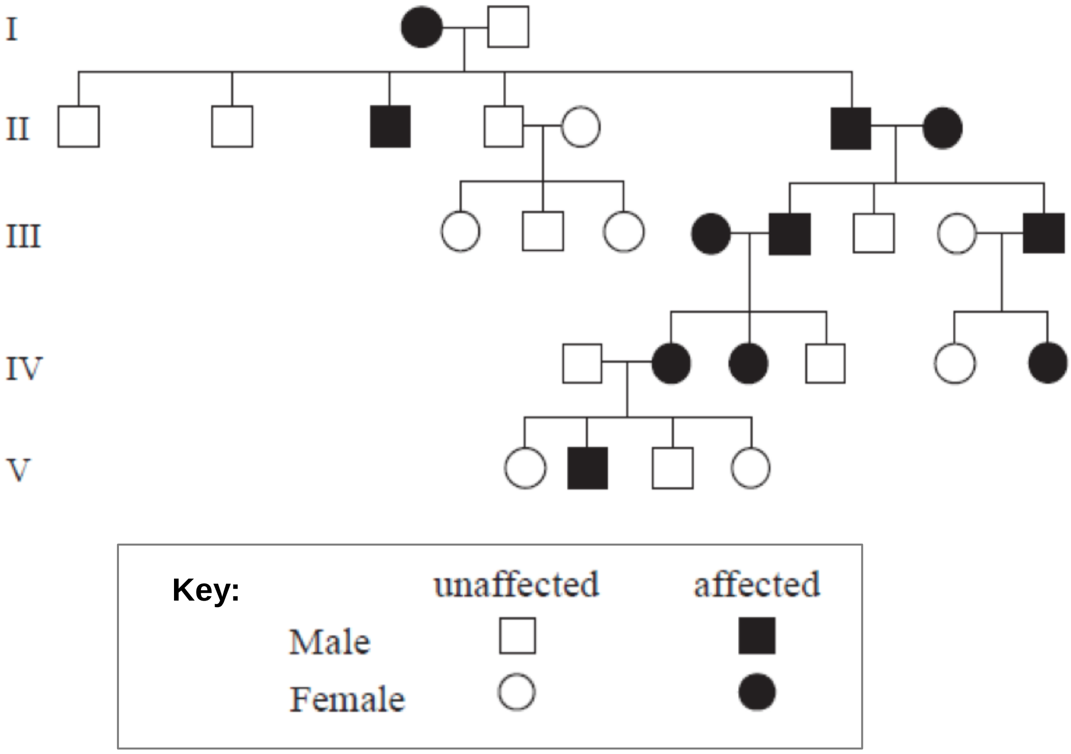
15. In rats, the gene for coat mottling has two alleles. The allele of a gene for 'mottled' coat (M) and the recessive allele (m) for 'normal' coat are sex-linked. The gene for whisker length also has two alleles. The allele of a gene for 'long' whiskers (W) and the recessive allele (w) for 'short' whiskers are autosomal.

A male rat with a normal coat and long whiskers was mated with a female with a mottled coat and long whiskers.

Given that the genotypes of both parents are unknown, which of the following scenarios is **not** possible?

	Genotype of parent	**Result**
A	Female is homozygous dominant at gene loci for coat mottling.	Female offspring with mottled coat and short whiskers may be observed.
B	Male is heterozygous at gene loci for whisker length.	Male offspring with mottled coat and short whiskers may be observed.
C	Female is homozygous dominant at gene loci for both coat mottling and whisker length.	Male offspring with normal coat and long whiskers may be observed.
D	Female is heterozygous at gene loci for both coat mottling and whisker length.	Female offspring with normal coat and long whiskers may be observed.

16. The pedigree chart below shows the inheritance of a genetic disease in a family.



What is the nature of the allele that causes this disease?

A autosomal dominant

B sex-linked dominant

C autosomal recessive

D sex-linked recessive

17. Fig. 17 below shows a human karyotype.

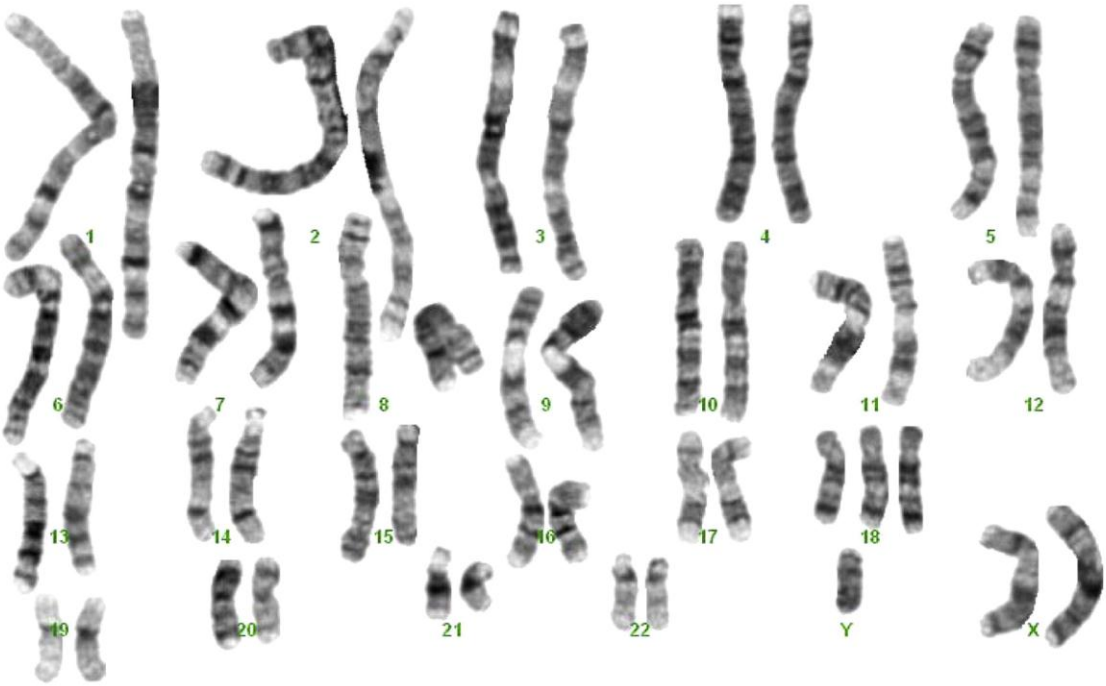


Fig. 17

Which of the following is evident from **Fig. 17**?

A The karyotype indicates a female patient with a single chromosomal aberration.

B The karyotype indicates a male patient with a single chromosomal duplication.

C The karyotype indicates a female patient with two chromosomal duplications.

D The karyotype indicates a male patient with two chromosomal aberrations.

18. In an experiment, chloroplast extracts are first treated with a chemical that 'snatches' away the electron that is accepted by the electron acceptor in photosystem I. The extracts are then treated with 2 hours of light and are provided with ample carbon dioxide and water.

Which of the following **correctly** shows the products that are formed after the experiment?

	O_2	**ATP**	**Reduced NADP**	**Glucose**
A	+	+	–	–
B	–	+	+	+
C	+	–	–	–
D	–	–	+	–

Key: (+) = present, (–) = absent

19. Fig. 19 below illustrates several metabolic processes that occur within the body. **Table 19** contains three statements that describe some of these processes.

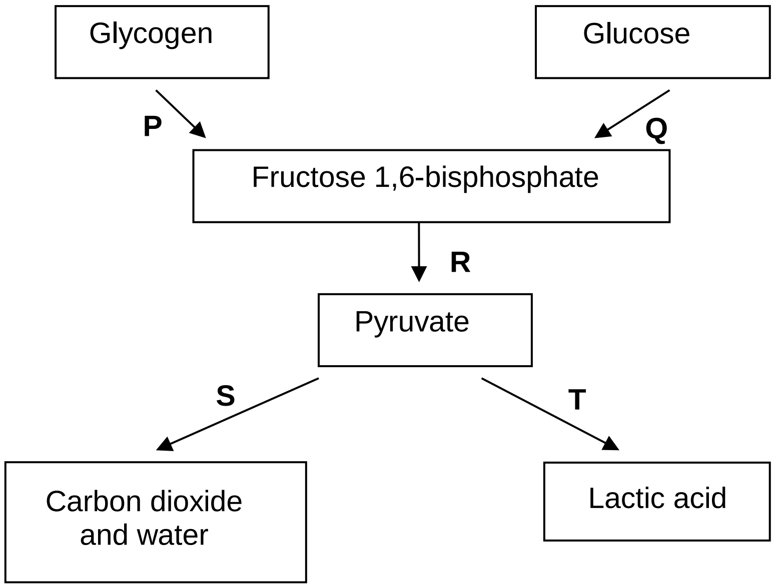


Fig. 19

Statement 1	NAD^+ is regenerated without the use of the electron transport system.
Statement 2	ATP is synthesised via substrate-level phosphorylation.
Statement 3	It can occur under anaerobic conditions.

Table 19

Which of the following correctly relates processes **P, Q, R, S, T** to statements **1, 2** and **3**?

	Statement 1	**Statement 2**	**Statement 3**
A	R	S	T
B	T	R	P
C	S	Q	S
D	Q	S	R

20. Which of the following correctly indicates a similarity and difference between the electron transport chains involved in photosynthesis and respiration?

	Similarity	Difference	
		Photosynthesis	**Respiration**
A	Channel proteins are components of both electron transport chains.	ATP synthesis does not occur.	ATP synthesis occurs.
B	Regeneration of electron carriers occurs at both electron transport chains.	$NADP^+$ is regenerated.	NAD^+ is regenerated.
C	Water is a direct source of electrons for both electron transport chains.	Cytochrome c oxidase not involved.	Cytochrome c oxidase involved.
D	Membrane integrity is important for the proper functioning of both electron transport chains.	H^+ is pumped inwards into thylakoid space.	H^+ is pumped outwards into intermembrane space.

21. Which of the following describes the process of natural selection?

A Different rates of reproductive success of different genotypes.

B Spontaneous occurrence of advantageous mutations.

C Change from simple to more complex organisms over time.

D Change in the size of population over time.

22. The Guinea baboon and King colobus are two of the many Old World monkeys that exist.

Guinea baboon

King colobus

The classification of these monkeys is shown below.

- Animalia
  - Chordata
    - Mammalia
      - Primates
        - Cercopithecidae (sub: Cercopithecinae)
          - Papio
            - *Papio papio* (Guinea baboon)

- Animalia
  - Chordata
    - Mammalia
      - Primates
        - Cercopithecidae (sub: Colobinae)
          - Colobus
            - *Colobus polykomos* (King colobus)

Which of the following conclusions is consistent with the information above?

A Both belong to the same class but different families.

B Both belong to the same genus but different classes.

C Both belong to the same family but different genera.

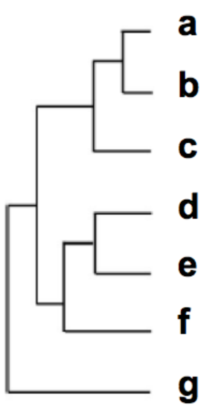
D Both belong to the same species but different orders.

23. The table shows the number of estimated nucleotide substitutions that have occurred since the divergence of seven species **a** to **g**.

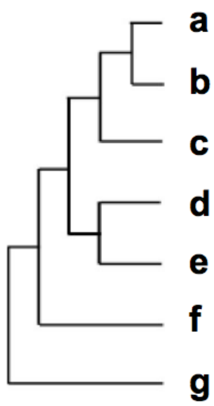
	b	**c**	**d**	**e**	**f**	**g**
a	39	72	128	126	159	269
b		81	130	128	158	268
c			129	127	157	267
d				56	154	271
e					151	268
f						273

Which of the following phylogenetic trees best shows the relationship among these seven species?

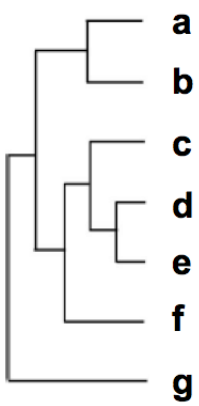
A



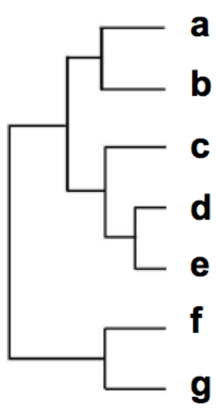
B



C



D



24. Which of the following statements regarding restriction enzymes is **incorrect**?

A Host DNA is prevented from digestion by restriction enzymes due to specific methylation of DNA.

B A restriction enzyme can create either sticky or blunt ends depending on the chemical conditions present.

C Cleavage by a restriction enzyme involves the breaking of two covalent phosphodiester bonds.

D Restriction enzymes protect bacteria against invading viruses by preventing the replication of viral DNA.

25. Which of the following does **not** occur during the polymerase chain reaction?

A Synthesis of a complementary primer

B Separation of parental DNA strands

C Formation of strong covalent bonds

D Involvement of inorganic enzyme co-factors

26. Which of the following is an outcome of the Human Genome Project that has ethical implications?

A Screening of the genetic make-up of newborn infants for susceptibility to certain key diseases.

B Creation of customised medicines that are potentially more expensive to produce than traditional drugs.

C Consideration of a suspect's genetic pre-disposition to violent behaviour in criminal trials.

D The free availability and accessibility of the complete sequence of the human genome on the Internet.

27. Totipotency is demonstrated when

A an embryonic stem cell divides and differentiates.

B cancer cells give rise to heterogeneous cell types.

C an isolated plant cell develops into a normal adult plant.

D a hematopoietic stem cell differentiates into a lymphocyte.

28. Which of the following methods can be used to introduce a foreign gene into a plant cell?

1. By *Agrobacterium*-mediated transfection
2. By microinjection of naked DNA
3. Using a phage delivery vector
4. Using a gene gun

A 1 and 3 only.

B 1, 2 and 4 only.

C 1, 3 and 4 only.

D All of the above.

29. There is concern regarding the escape of GM salmon into the wild because they

A may replace wild salmon populations.

B may not survive as well in the wild.

C are prone to further mutations.

D will produce sterile offspring with wild salmon.

30. Which of the following is **not** a possible concern of cultivating *Bt* corn?

A Toxic effects of *Bt* on non-target insects e.g. monarch butterfly larvae could have predictable ecological consequences.

B Transfer of selection marker to bacteria that reside in the human gut thereby conferring upon the bacteria antibiotic resistance.

C Spread of the *Bt* gene from cultivated corn to wild relative which could then lead to the loss of biodiversity.

D Transfer of *Bt* gene to the pests thereby increasing their resistance to the *Bt* toxin